ANSWERS

Year 10
General
Test 4
2019

Baldivis
Secondary College

Name: ______________________ Class: ______ Time: 40 minutes Total Mark:____/36

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

Resources: 1 A4 page of notes (1 side), Calculator allowed.

Question 1 **5 marks**

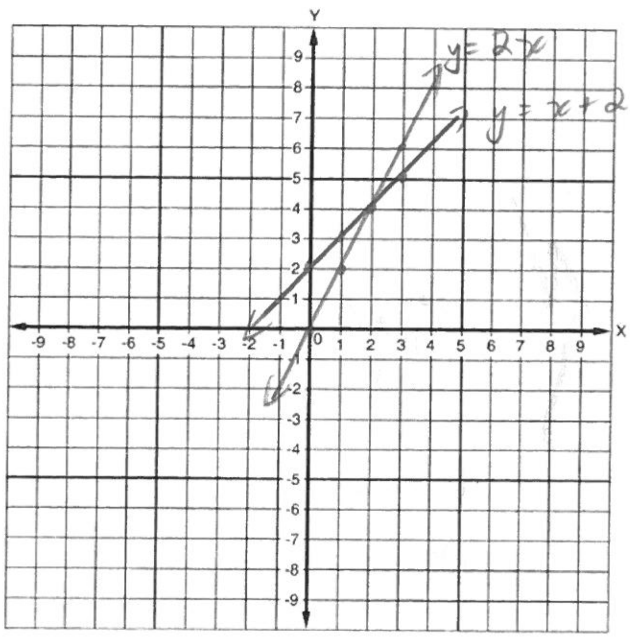
Plot the following linear graphs for the given range of x – values on the same Cartesian plane:

a) $y = x + 2$

x	0	1	2	3
y	2	3	4	5

b) $y = 2x$

x	0	1	2	3
y	0	2	4	6



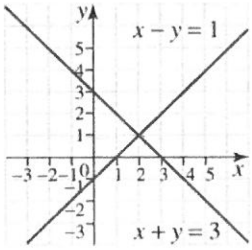
c) Use these graphs to determine the point of intersection and, hence, the solution of the simultaneous equations.

Solution is (2,4)

Question 2 — 2 marks

Use the graphs below of the given simultaneous equations to write the point of intersection and, hence, the solution of the simultaneous equation.

Equation 1: $x - y = 1$
Equation 2: $x + y = 3$



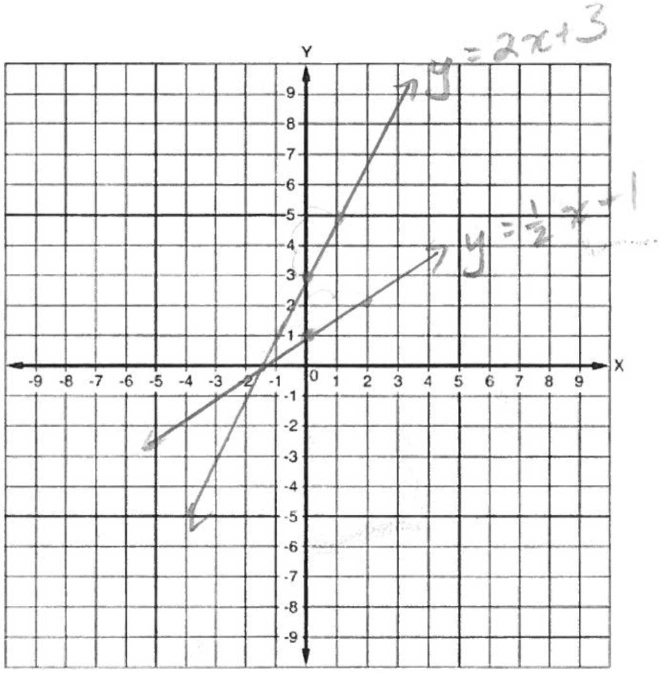
(2, 1)

Question 3 — 5 marks

Solve the following pair of simulations equations using a graphical method

a. $y = \overset{m}{2}x + \overset{c}{3}$ y-intercept = (0, 3) gradient = 2 = $\frac{2}{1}$ $\frac{\text{rise}}{\text{run}}$

b. $y = \frac{1}{2}x + 1$ y intercept = (0, 1) gradient = $\frac{1}{2}$ $\frac{\text{rise}}{\text{run}}$



c. Use these graphs to determine the point of intersection and, hence, the solution of the simultaneous equations.

(−1.5, 0) approx

Question 4	10 marks

Solve each of the following equations (that is, find the value for the pronumeral that makes each statement true).

a) $x + 2 = 8$

$-2 \quad -2$

$x = 6$

b) $\frac{6k}{6} = \frac{36}{6}$

$k = 6$

c) $\frac{w}{5_{\times 5}} = -9 \times 5$

$w = -45$

d) $5a + 6 = 26$

$-6 \quad -6$

$\frac{5a}{5} = \frac{20}{5}$

$a = 4.$

e) $\frac{f}{4} + 6 = 16$

$-6 \quad -6$

$\frac{f}{4_{\times 4}} = 10^{\times 4}$

$f = 40$

f) $\frac{x}{2_{\times 10}} - \frac{3x^{\times 4}}{5_{\times 4}} = \frac{1}{4_{\times 5}}$

$\frac{10x}{20} - \frac{12x}{20} = \frac{5}{20}$

$10x - 12x = 5$

$\frac{-2x}{-2} = \frac{5}{-2}$

$x = -2\frac{1}{2}$

Question 5	4 marks

Find the gradient of each of the following pairs of lines and determine if they are perpendicular or parallel.

a) $y = 2x + 5$ and $y = 2x + 6$

Gradients: 2 Are these lines parallel/perpendicular (circle the correct answer)

b) $y = \frac{1}{3}x + 1$ and $y = -3x + 2$

Gradients: $\frac{1}{3}$ + -3 Are these lines parallel/perpendicular (circle the correct answer)

$m_1 \times m_2 = -1$

$\frac{1}{3} \times -3 = -1$

Question 6	6 marks

Solve the following pair of simultaneous equations using an algebraic method:

$$y = 2x$$
$$x + y = 18$$

$x + 2x = 18$

$\frac{3x}{3} = \frac{18}{3}$

$x = 6$

sub $x = 6$ into $y = 2x$

$y = 2x$

$y = 2 \times 6$

$y = 12$

solution is $x = 6$, $y = 12$

Question 7	4 marks

Look at the 5 equations below

i) $y = 3x + 3$
ii) $3x + y = 3$
iii) $3x - y = 3$
iv) $3x + 3y = 2$
v) $3y = x + 3$

i) $y = 3x + 3$

ii) $3x + y = 3$
$-3x \quad -3x$
$y = 3 - 3x$
$y = -3x + 3$

iii) $3x - y = 3$
$-3x \quad -3x$
$\frac{-y}{-1} = \frac{3 - 3x}{-1}$
$y = -3 + 3x$
$y = 3x - 3$

iv) $\frac{3x}{3} + \frac{3y}{3} = \frac{2}{3}$
$x + y = \frac{2}{3}$
$-x \quad -x$
$y = \frac{2}{3} - x$
$y = -x + \frac{2}{3}$

v) $\frac{3y}{3} = \frac{x + 3}{3}$
$y = \frac{1}{3}x + 1$

a) Which two of the lines are parallel?

$y = 3x + 3$ (i) & $3x - y = 3$ (iii)

b) Which two of the lines are perpendicular?

$3x + y = 3$ (ii) & $3y = x + 3$ (v)

END OF TEST